

Problem Solving and Programming
PSP – CS105
Assignment-4
Deadline: 1st December 2025 till 15:30

[Q1] Write a C++ program that gets a positive integer from the user. Your program should then find out whether the given number is a palindrome or not. Your program should also count the total number of odd digits and display if the total count of odd numbers is a palindrome. Your program should be repeated if the user wants to continue. In case, user presses Y or y, the program should continue until the user keeps on pressing Y or y. If the user presses N or n, the program should immediately stop. In case any other letter is entered, your program should display “Invalid” and give the user three chances to make sure that the user enters a valid option(Y/y/N/n). If a valid option is not pressed, your program should immediately terminate.

Hint: A palindrome number is a sequence of digits that is read the same forward and backward. For example, 1221, 353, and 6788876 are all palindrome numbers because when you read 1221 from right to left side it will be the same number 1221. Numbers like 354, 63, 8975 are not palindrome numbers, as in 354, if read from right to left, will not give the same number (the number here is 453 when read from right to left, which is not the same or equal to 354)

[Q2] Write a C++ program that first gets a positive integer from the user. Your program should then scan all digits one by one and process each digit to find if it is a prime number. Your program should count the total number of prime digits and display it. Your program should then ask the user if the user wants to continue, and repeat this procedure for any other positive user-given number. In case, user presses Y or y, the program should continue until the user keeps pressing Y or y. If the user presses N or n, the program should immediately stop. In case any other letter is entered, your program should display “Invalid” and give the user three chances to make sure that the user enters a valid option(Y/y/N/n). If a valid option is not pressed, your program should immediately terminate.

(Hint: For instance, if a user gives a number 38545, it should display that “38545 has 3 prime digits”).

[Q3] Pascal’s triangle is a triangular array of the binomial coefficients. Write a C++ program that takes an integer value n as input and prints the first n lines of Pascal’s triangle. The following are the first 6 rows of Pascal’s triangle.

```
1
1   1
1   2   1
1   3   3   1
1   4   6   4   1
1   5  10  10   5   1
```

Instructions:

1. There is zero tolerance for cheating. Cheating will be penalized.
2. Submit your complete source code. Place your source code in a folder and name the folder as your_name_A4-PSP-F25. Compress the folder and submit the zip file in the submission folder, referred to as PSP-A4-F25, on fs.